

sunRunner3D MHD Simulations of Solar Cycle 25 Geomagnetic Storms: Heliospheric Propagation and In-Situ Comparison of Extreme CME Events



E. E. Díaz-Figueroa^{1,2}, J. J. González-Avilés¹

¹Escuela Nacional de Estudios Superiores, Unidad Morelia, Universidad Nacional Autónoma de México ²Escuela Normal Superior de México

1. Abstract

Intense geomagnetic storms are among the space-weather phenomena with the greatest potential impact on technological infrastructure. We study highly energetic coronal mass ejections (CMEs) of Solar Cycle 25 with three-dimensional MHD simulations, using **sunRunner3D** (SR3D) to propagate them through the inner heliosphere, 0.14–1.1 au. Events are drawn from a catalogue filtered by **geomagnetic response** — NOAA G4 (severe) or above, i.e. $Kp \geq 8$ — with low- to mid-latitude source regions and Earth-directed longitudes. Simulated density, radial velocity and radial magnetic field are extracted at Earth's position and compared against in-situ L1 measurements, *averaged to the model output cadence* so that both series share the same effective resolution. The aim is to characterise the interplanetary conditions driving extreme storms — arrival profiles, transit times and upstream wind structure — towards improved forecasting tools.

Scope of this poster. Of that catalogue, a **single event** is presented here: the CME of **11 November 2025** (AR 14274), the **strongest X-class flare of the year** (X5.1, peak at 10:04 UT) and driver of the **G4 severe geomagnetic storm** of 12 November ($Kp \geq 8$ —, $Dst_{\min} \approx -217$ nT). The remaining events of the catalogue are the subject of ongoing work.

2. Numerical model

sunRunner3D (SR3D), built on the **PLUTO** framework, solves the ideal MHD equations in conservative form in an inertial spherical frame (r, θ, ϕ):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (E1)$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v} - \mathbf{B} \mathbf{B} + p_t \mathbf{I}) = 0, \quad (E2)$$

$$\frac{\partial E}{\partial t} + \nabla \cdot ((E + p_t) \mathbf{v} - \mathbf{B}(\mathbf{v} \cdot \mathbf{B})) = 0, \quad (E3)$$

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \cdot (\mathbf{v} \mathbf{B} - \mathbf{B} \mathbf{v}) = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad (E4)$$

Domain	0.14–1.1 au, full 4π
Grid	$70 \times 80 \times 128$ (r, θ, ϕ)
Solver	HLLC, 2nd-order (minmod) + RK2
$\nabla \cdot \mathbf{B}$ control	Powell 8-wave
Inner boundary	CORHEL/MAS coronal solution at 0.14 au
Solar rotation	boundary rotated at $\Omega_c = 2\pi/25.38 \text{ d}^{-1}$

3. Event and CME injection

The **2025-11-11T10:04 CME** (AR 14274, X5.1 flare) is injected at the inner boundary as a **geometric cone** — a hydrodynamic density/velocity pulse — guided by the NASA/CCMC **DONKI** catalogue (SOHO/LASCO + STEREO/COR2).

Parameter	Value
Carrington rotation	2304
Run window / cadence	11-06 – 11-16 UT, 10.0 h
Injection start / end	2025-11-11, 10:13 / 14:28 UT
V_r (km s ⁻¹)	1856
N_p (cm ⁻³)	1800
Latitude / Carrington longitude (°)	+24 / 267.4
Cone half-width (°)	45

N_p is not constrained by coronagraph imaging; it is estimated from the n – v anticorrelation [2].

4. Results

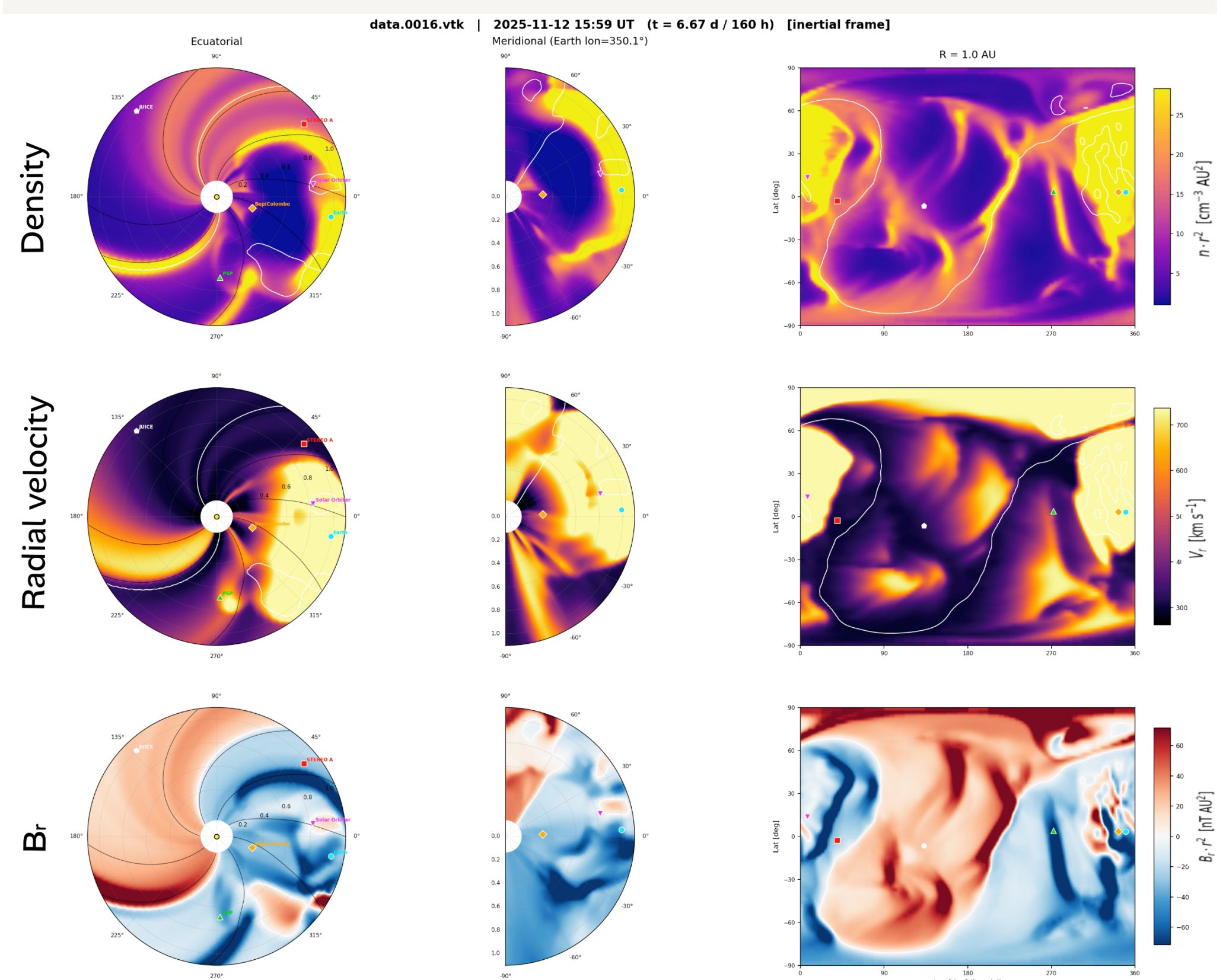


Figure 1: Arrival of the CME front at 1 AU on November 12, 2025, at 15:59 UT: first frame in which the front reaches Earth's position..

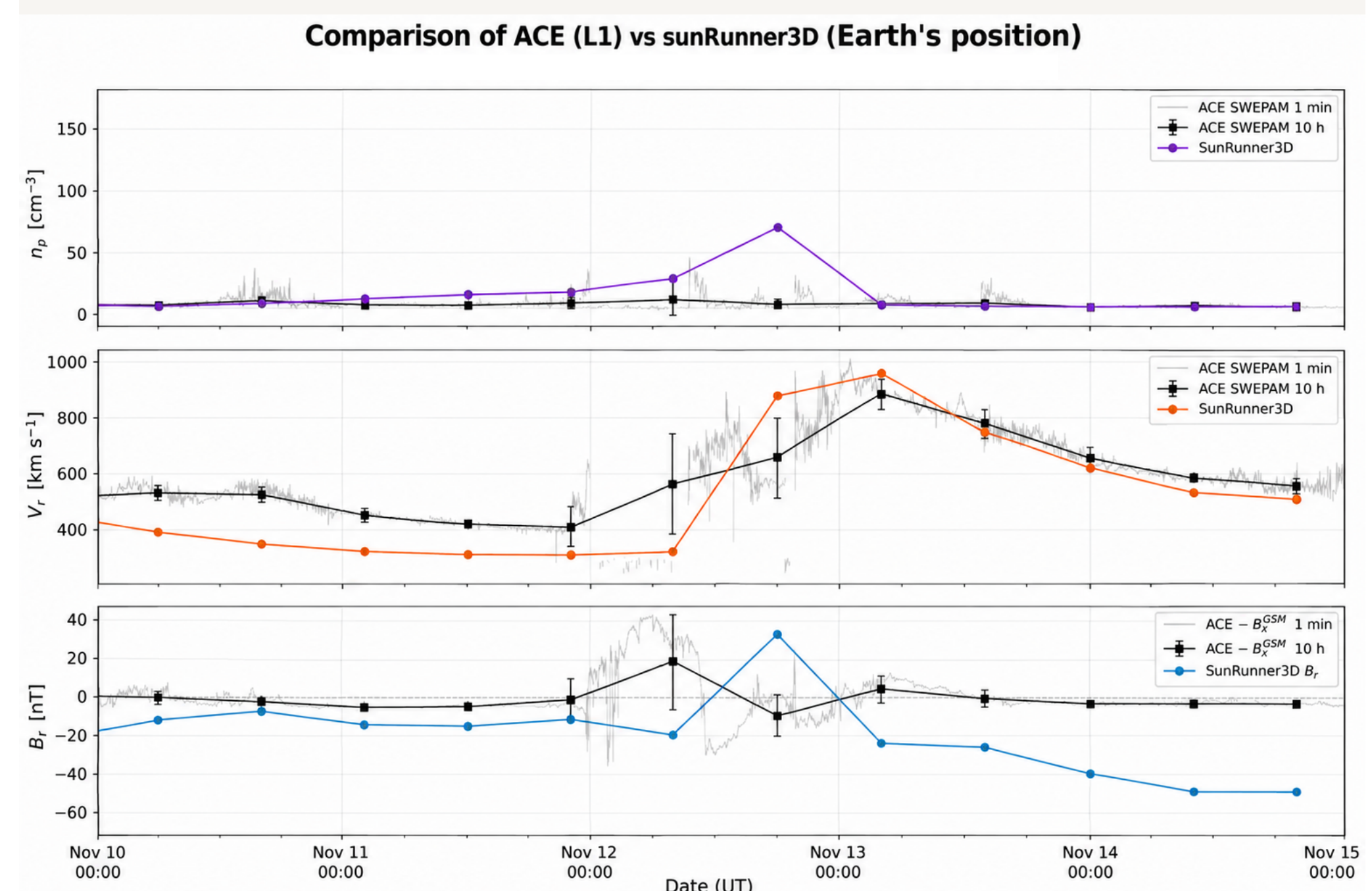


Figure 2: In-situ comparison. Grey: ACE at 1 min. Black: ACE averaged to the model cadence. Colour: SR3D.

5. Conclusions

- Arrival and peak V_r captured** within the 10-h cadence and to $\sim 8\%$ — SR3D is a viable arrival-time forecast tool.
- n_p and B_r overpredicted** ($\sim 7\times$, $\sim 2\times$); a flux-rope injection is the next step.